



APEX ENGINEERING GROUP
Engineering Flow, Designing Solutions

CFD Portfolio

Company Profile

*We are **APEX ENGINEERING GROUP**, an engineering consultancy specializing in Computational Fluid Dynamics (CFD), thermal analysis, and flow optimization.*

We support clients in improving product performance, identifying inefficiencies, and making data-driven engineering decisions using advanced simulation tools such as ANSYS Fluent.

Attached are projects demonstrating our technical approach, analysis capabilities, and the value we deliver to clients.

We would be pleased to support your project with accurate, high-quality engineering solutions.

Best

regards,

APEX ENGINEERING GROUP

Project 1

CFD Analysis of Quenching Chamber

Industry

Polymer Processing Industry

Problem Statement

The client faced non-uniform cooling in a quenching chamber, leading to inconsistent fiber properties and potential product defects.

Objective

To analyze airflow distribution and thermal behavior within the chamber and identify opportunities to improve cooling uniformity.

Approach & Methodology

- CFD simulation using ANSYS Fluent
- Laminar, incompressible flow
- Thermal analysis of stationary fiber
- Evaluation of velocity, pressure, and temperature fields

Results

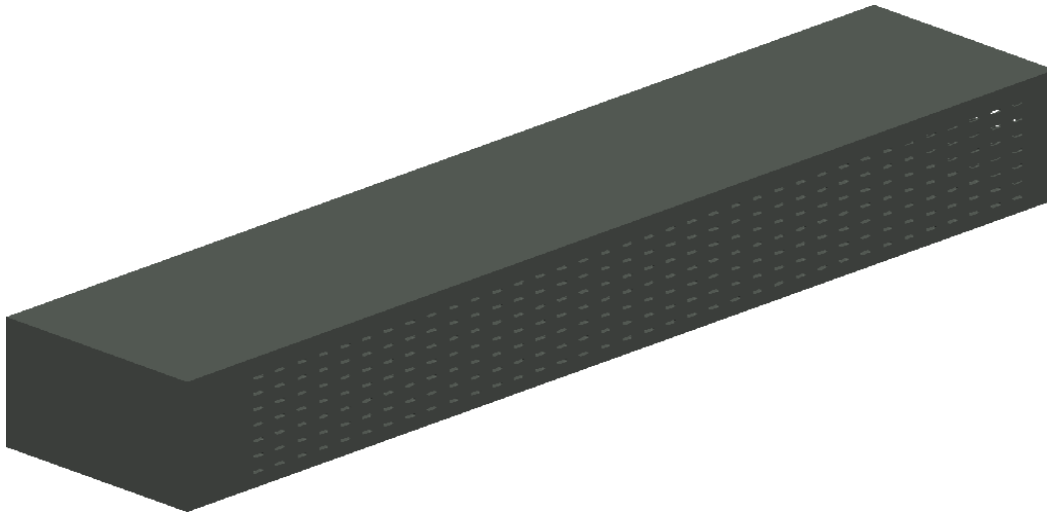


Figure 1 geometry

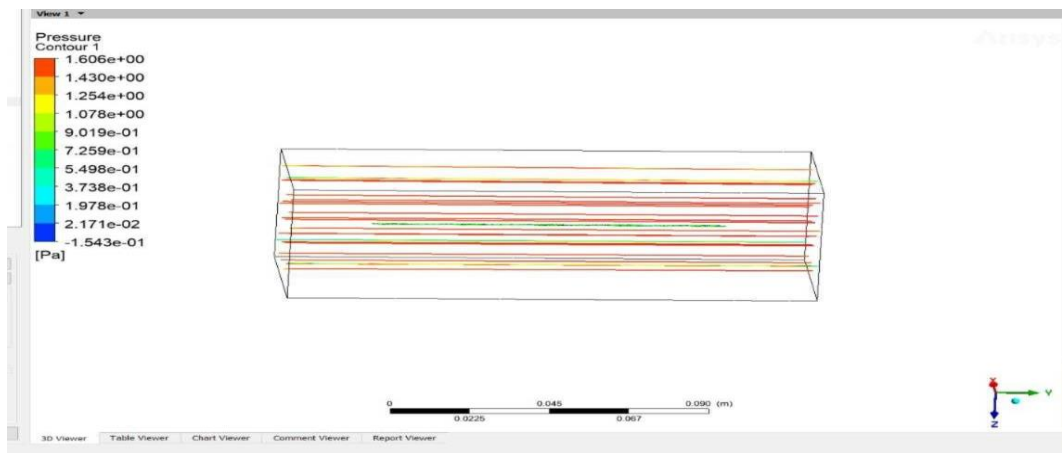


Figure 2 Pressure contour

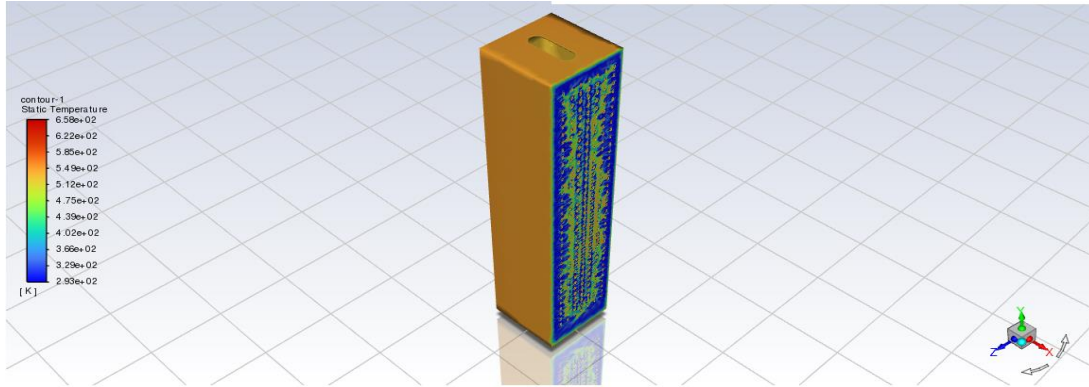


Figure 3 Temp contour

Key Findings

- Stable and predominantly axial airflow
- Wake region behind fiber reduces cooling efficiency
- Non-uniform temperature distribution
- Low turbulence limits heat transfer

Results Impact

- Identified critical cooling inefficiencies
- Provided actionable design improvements
- Reduced risk of product defects
- Enabled better engineering decisions

Conclusion

The system exhibits stable airflow but shows limitations in cooling uniformity due to flow behavior around the fiber. Optimizing airflow design and enhancing mixing can improve thermal performance, reduce non-uniformity, and help minimize material defects, leading to improved product quality and reduced production losses.

Project 2

CFD Thermal Analysis of Finned Tube Heat Exchanger

Industry

HVAC / Heat Transfer Systems

Problem Statement

The client required thermal performance evaluation of a finned tube heat exchanger to study temperature distribution and heat transfer efficiency across the coil.

Objective

To analyze the temperature variation along the tube network and evaluate thermal behavior of the heat exchanger under operating conditions.

Approach & Methodology

- CFD simulation using ANSYS Fluent
- Steady-state thermal analysis
- Heat transfer evaluation through finned tubes
- Temperature contour visualization and performance assessment

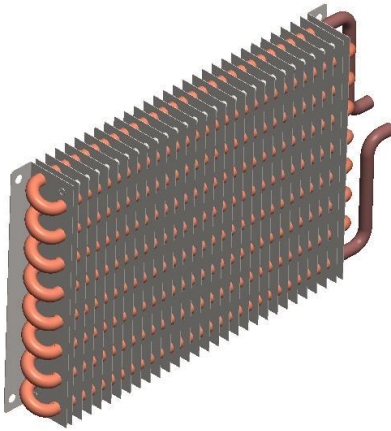


Figure 1 Geometry of finned tube heat exchanger

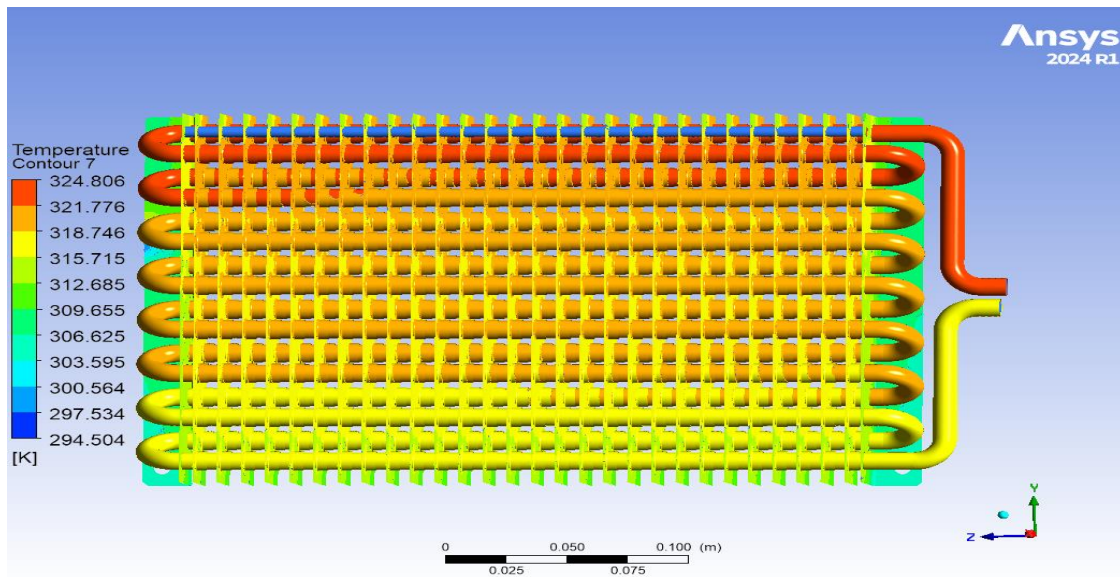


Figure 2 Temperature contour showing thermal distribution

Key Findings

- Gradual temperature reduction observed across the coil
- Heat transfer through fins improved thermal dissipation
- Temperature contours identified thermal gradients within the system
- Simulation validated effective thermal distribution behavior

Results Impact

- Assisted in evaluating heat exchanger efficiency
- Helped identify thermal performance characteristics
- Supported design optimization for improved cooling/heating performance
- Reduced development and testing effort through virtual analysis

Conclusion

The CFD analysis successfully evaluated the thermal performance of the finned tube heat exchanger. The temperature contours demonstrated effective heat transfer throughout the system while highlighting areas of thermal variation. The study provides valuable insight for optimizing heat exchanger performance and improving overall system efficiency.

Project 3

Formula 1 Front Wing Aerodynamic CFD Analysis

Project Overview

This CFD project evaluates the aerodynamic performance of a multi-element front wing using different flap angle and slot gap configurations. The study focuses on airflow acceleration, streamline behavior, and aerodynamic efficiency.

Industry

Automotive Aerodynamics / Motorsport CFD

Problem Statement

The client required aerodynamic performance evaluation of a multi-element front wing to study airflow behavior, drag characteristics, and the effect of flap angle and slot gap configurations on aerodynamic efficiency.

Objective

To analyze airflow acceleration, streamline interaction, pressure distribution, and aerodynamic behavior of the front wing under different design configurations in order to identify an optimized aerodynamic setup.

Approach & Methodology

- 3D front wing geometry designed and prepared for CFD analysis
- Computational domain and enclosure generation
- CFD simulation using ANSYS Fluent
- External aerodynamic flow analysis
- Velocity contour and streamline visualization
- Evaluation of drag force and aerodynamic performance
- Comparative analysis of flap angle and slot gap configurations

Results

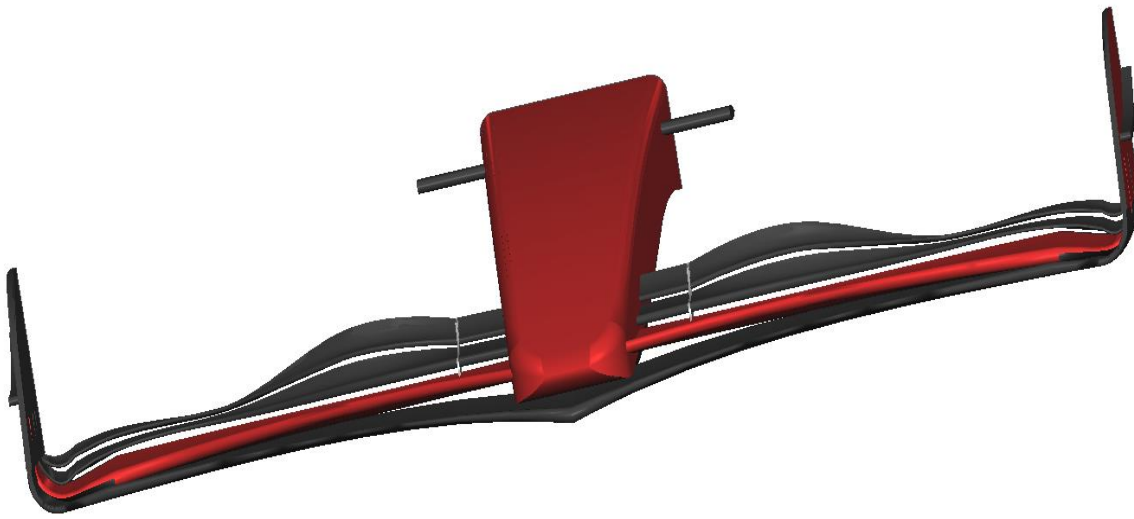


Figure 1 Front Wing Geometry

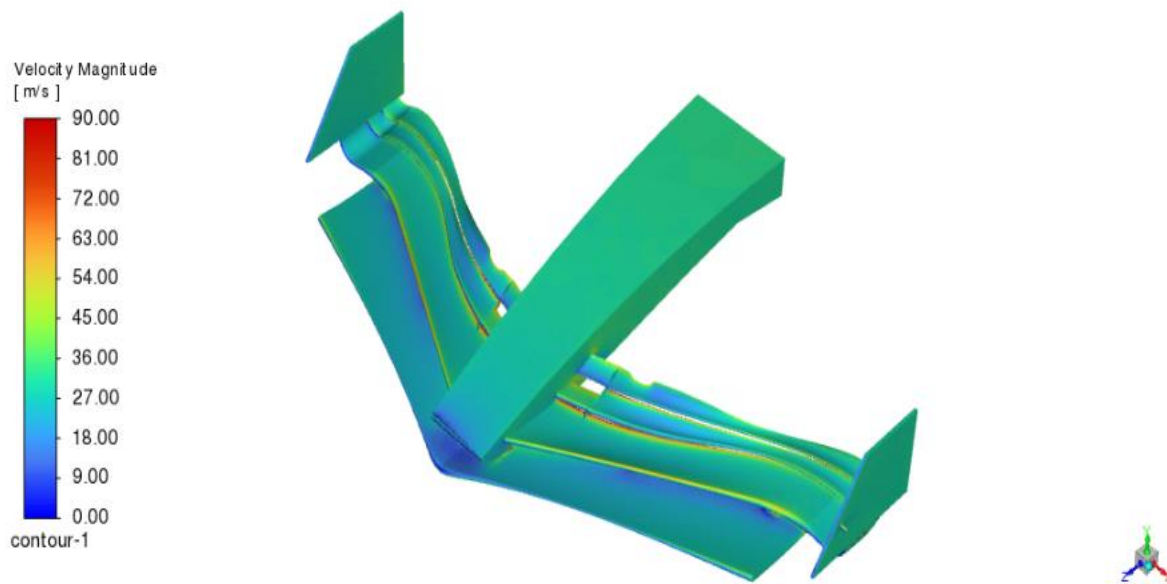


Figure 2 Velocity Contour Visualization

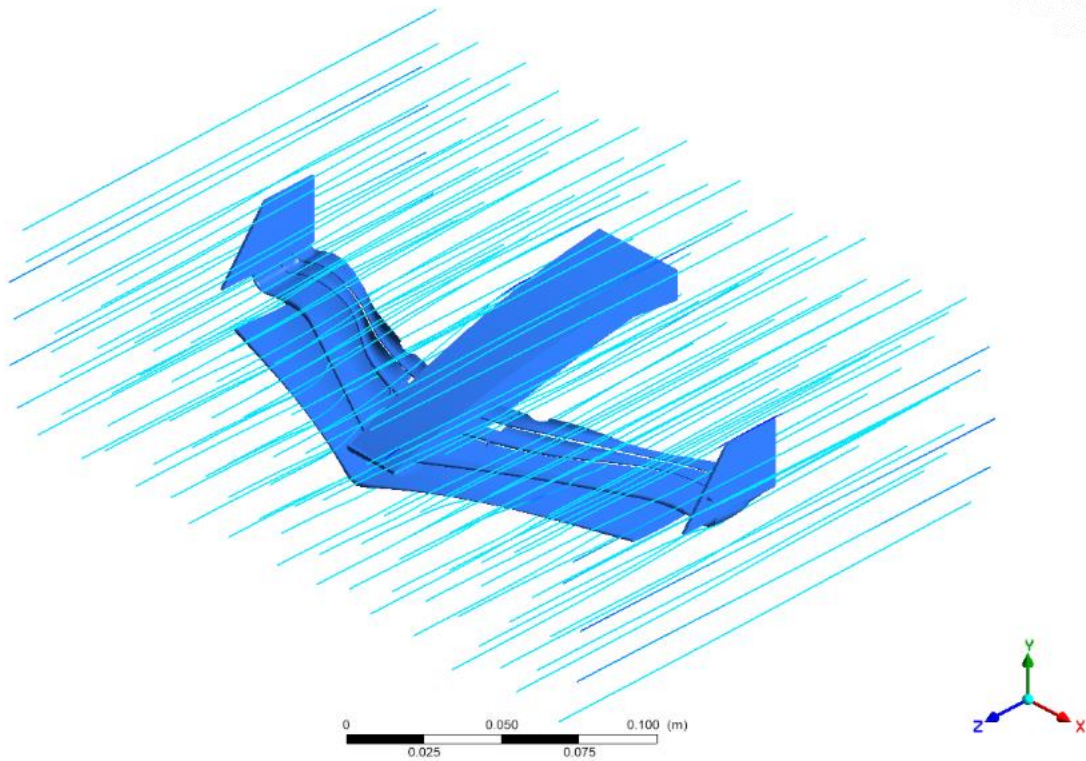


Figure 3 Velocity Streamline Analysis

Key Findings

- Airflow acceleration increased with optimized flap angle configuration
- Streamline interaction between wing elements significantly affected aerodynamic performance
- Smaller slot gaps improved flow energy transfer between elements
- Larger slot gaps resulted in weaker aerodynamic interaction and reduced effectiveness
- Balanced configurations produced improved aerodynamic efficiency and controlled flow behavior
- Drag and aerodynamic characteristics varied depending on flap angle and slot gap arrangement

Results Impact

- Assisted in evaluating aerodynamic efficiency of the front wing
- Helped identify optimized flap angle and slot gap configuration

- Supported aerodynamic design optimization before physical testing
- Best case produced approximately 9.4% improvement in lift performance, although drag force increased by approximately 11.8% compared to the baseline configuration due to stronger aerodynamic interaction and increased flow acceleration.

Conclusion

The CFD analysis successfully evaluated the aerodynamic performance of the multi-element front wing under different configurations. The study demonstrated how flap angle and slot gap significantly influence airflow acceleration, streamline interaction, and aerodynamic efficiency. The optimized configuration achieved balanced flow behavior with improved aerodynamic effectiveness, providing valuable insight for aerodynamic design optimization and performance enhancement.

Project 4

CFD Study of Battery Spacing and Airflow for Efficient Cooling of Li-ion Packs

Project Overview

- This project focuses on the thermal management of lithium-ion battery packs using Computational Fluid Dynamics (CFD). Different airflow configurations and cell spacings were analyzed to improve battery cooling performance, temperature uniformity, and airflow distribution inside the battery module.
- The study was performed using ANSYS Fluent by simulating airflow, heat transfer, pressure drop, and temperature distribution inside cylindrical battery arrangements.

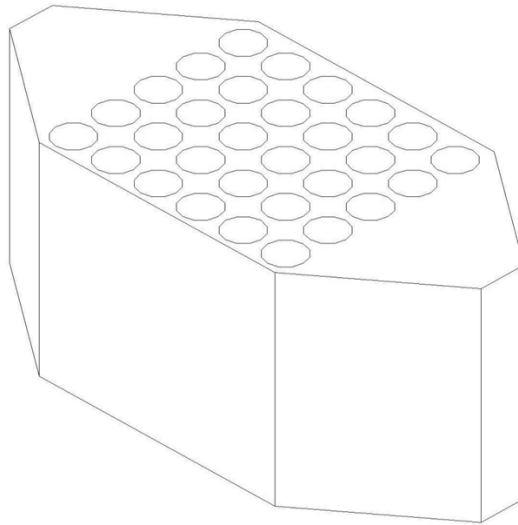


Figure 1: Isometric View of Lithium-ion battery pack of Straight Type

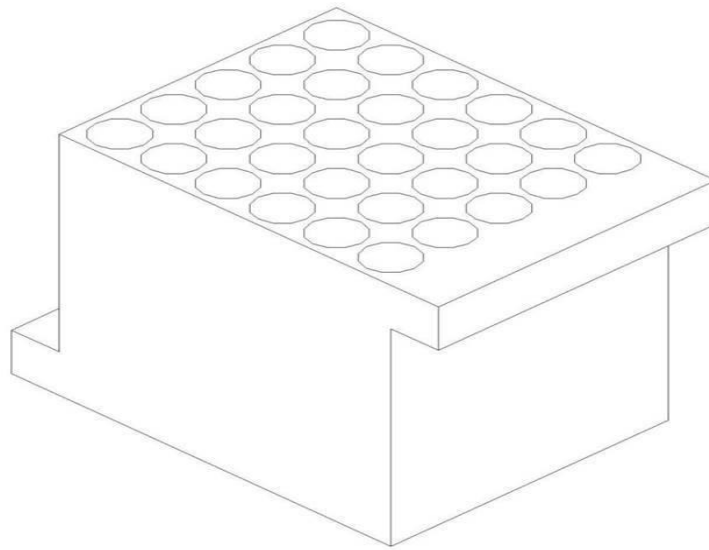


Figure 3: Isometric View of Z-Type Lithium-ion battery pack

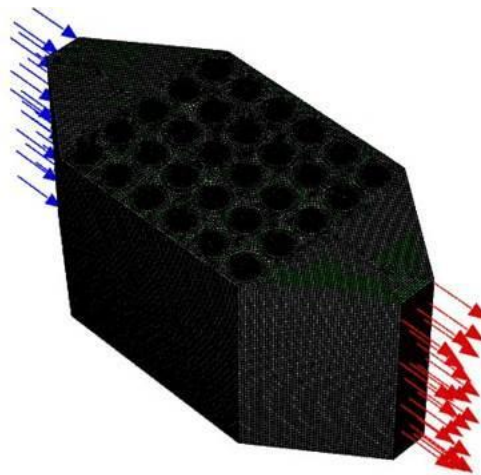


Figure 4: Isometric View of Lithium-ion battery pack meshing Straight Type

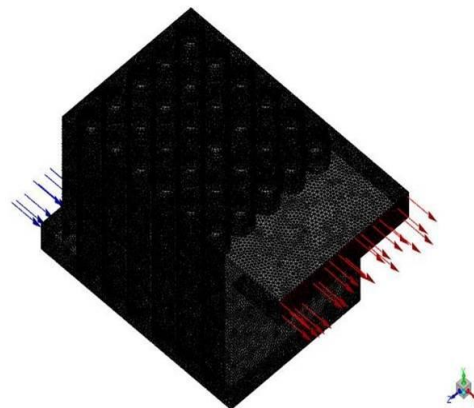


Figure 5: Top View of Lithium-ion Z-Type battery pack meshing

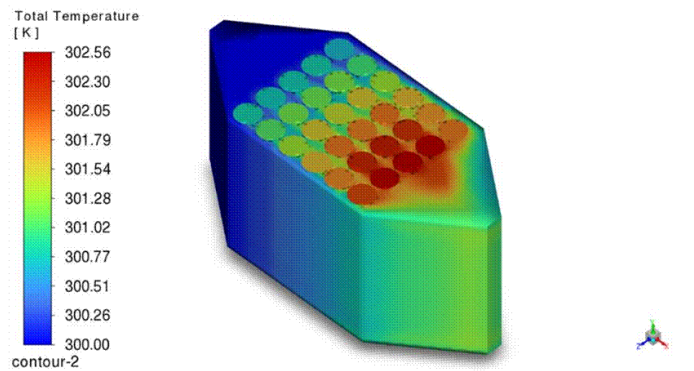


Figure 6.: Isometric View Temperature Contour of Straight Type

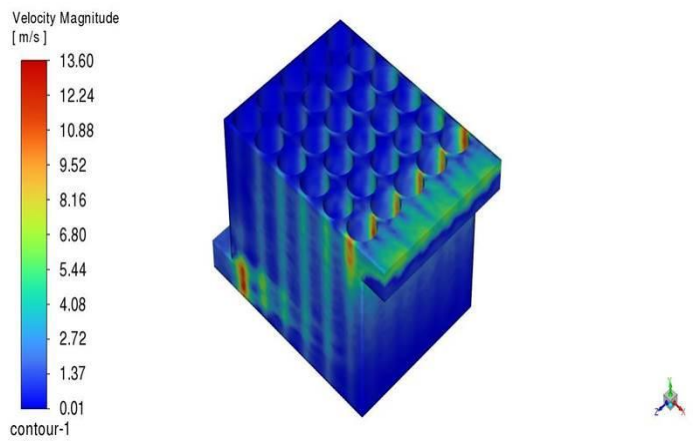


Figure 7: Isometric View of Velocity Contour of Z-Type

Software Used

- ANSYS Fluent
- ANSYS Meshing
- SolidWorks
- Microsoft Excel
- CFD Post

Services Offered

- ✓ CFD Simulation of Battery Cooling Systems
- ✓ Thermal Analysis of Lithium-Ion Battery Packs
- ✓ Airflow & Heat Transfer Analysis
- ✓ Meshing and Grid Independence Study
- ✓ Temperature, Velocity & Pressure Contours
- ✓ Engineering Reports & Documentation
- ✓ ANSYS Fluent Setup & Post-Processing
- ✓ Research & FYP Assistance

Project Features

- Straight-Type and Z-Type airflow configurations
- 1 mm, 2 mm, and 3 mm cell spacing analysis
- Mesh independence study
- Reynolds Number & Nusselt Number validation
- Temperature contour analysis
- Pressure drop evaluation
- Velocity distribution analysis
- Comparison with experimental/literature data

Key Results

- Improved airflow distribution with optimized spacing
- Reduced hotspot formation inside battery pack
- Lower pressure drop at larger spacing values
- Better thermal uniformity using Z-type airflow arrangement
- Validation performed using Reynolds and Nusselt number comparison

Skills Demonstrated

- CFD Modeling
- Thermal Management
- Numerical Simulation
- Engineering Analysis
- Heat Transfer
- Fluid Flow Analysis
- Technical Report Writing
- Data Visualization

Project 5

Aerodynamic Design and CFD Simulation of an Aircraft Wing Profile

Project Overview

This project demonstrates an industry-standard engineering workflow: designing a complex aircraft wing geometry in SolidWorks and performing a comprehensive Computational Fluid Dynamics (CFD) analysis in ANSYS Fluent. The objective was to evaluate external fluid flow dynamics, velocity profiles, and pressure distribution around the aerofoil, while ensuring mathematical validation through solution convergence mapping.

1. Geometric Modeling (SolidWorks)

- **CAD Generation:** The 3D geometry of the aircraft wing and aerodynamic profile was fully modeled using SolidWorks.
- **Design Strategy:** Applied parametric constraints to define the airfoil curves and internal structural spacing, ensuring a clean geometry with no self-intersections, making it perfectly optimized for fluid domain extraction.
- **Export:** Converted the native SolidWorks part into a neutral format (.STEP/.IGES) for seamless data transfer into the ANSYS environment.

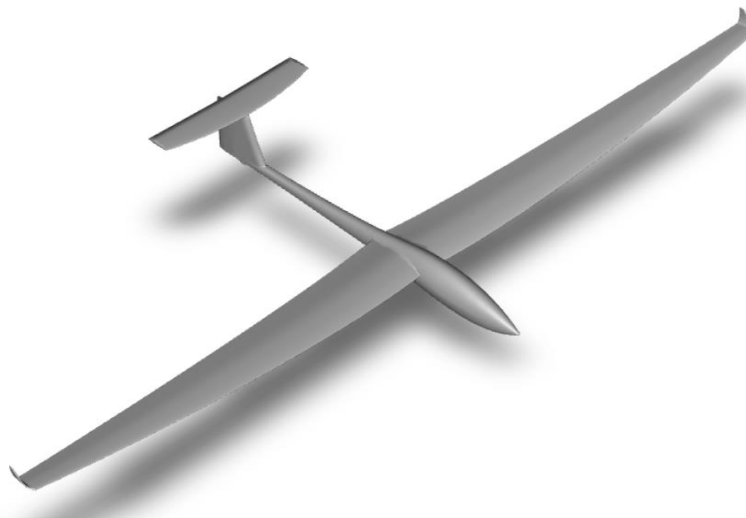


Figure 1: 3D CAD model of the aircraft wing surface.

2. Pre-Processing & Meshing (ANSYS Meshing)

- **Fluid Domain:** Imported the geometry into ANSYS and created an external virtual wind tunnel domain around the wing.
- **Grid Generation:** Used ANSYS Meshing to discretize the fluid domain.
- **Mesh Refinement:** Generated a high-quality unstructured mesh with critical refinement zones near the leading and trailing edges. Inflation layers were applied along the wing walls to accurately capture boundary layer gradients and turbulence effects.

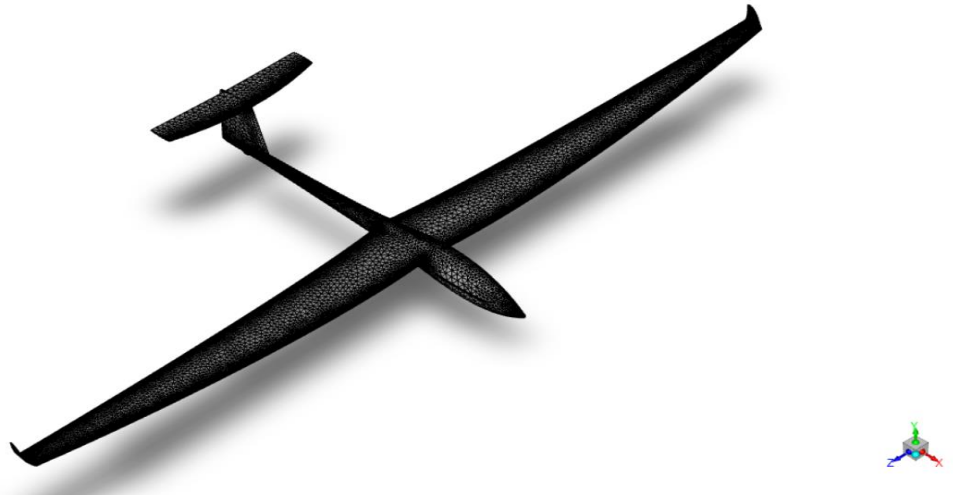


Figure 2: ANSYS computational mesh and fluid domain discretization

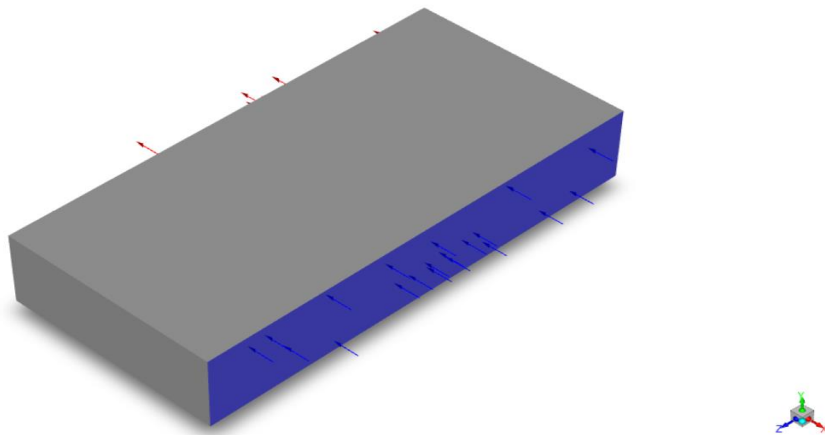


Figure 3: ANSYS Fluent residual monitors showing solution convergence

3. CFD Simulation Setup & Solution Convergence (ANSYS Fluent)

- **Solver Physics:** Pressure-based steady-state solver in **ANSYS Fluent** utilizing advanced viscous models (such as $k-\omega$ SST) to predict flow separation and wake formation.
- **Boundary Conditions:** Defined a uniform velocity inlet for airspeed, a pressure outlet for atmospheric exhaust, and a no-slip wall condition for the wing surface.
- **Solution Convergence & Validation:** Monitored the mass, momentum (x, y, z velocities), and turbulence energy residuals. As shown in the convergence history, all mathematical equations successfully stabilized and dropped below the strict convergence criteria, proving that the numerical solution is highly accurate and stable.

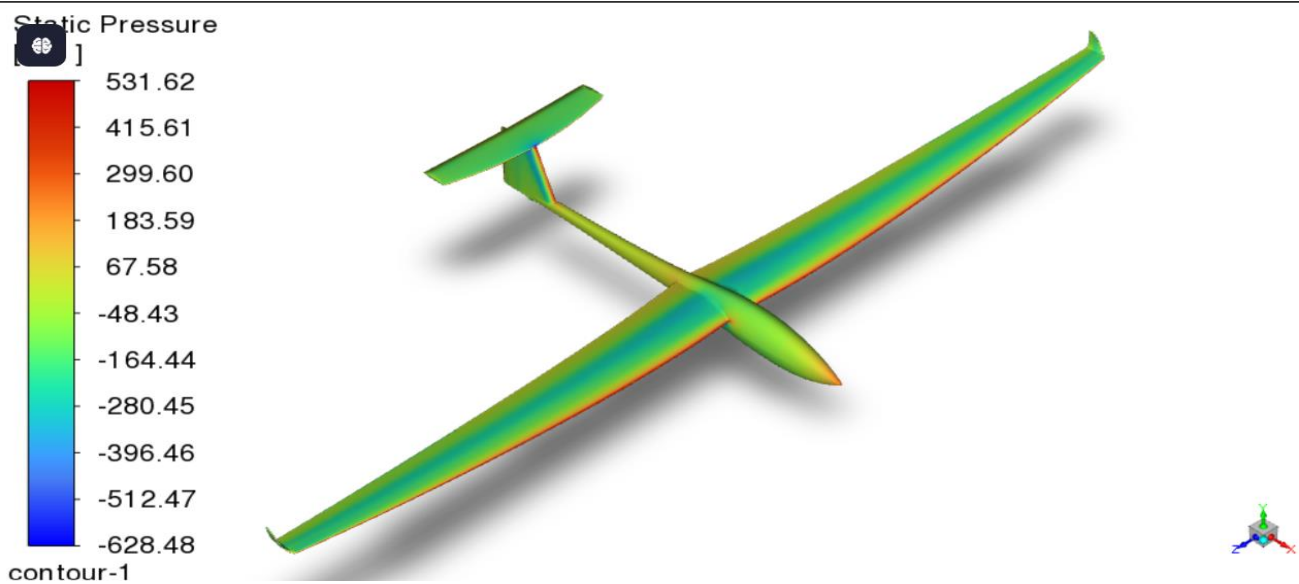


Figure 4: Total pressure contour demonstrating lift-generation zones

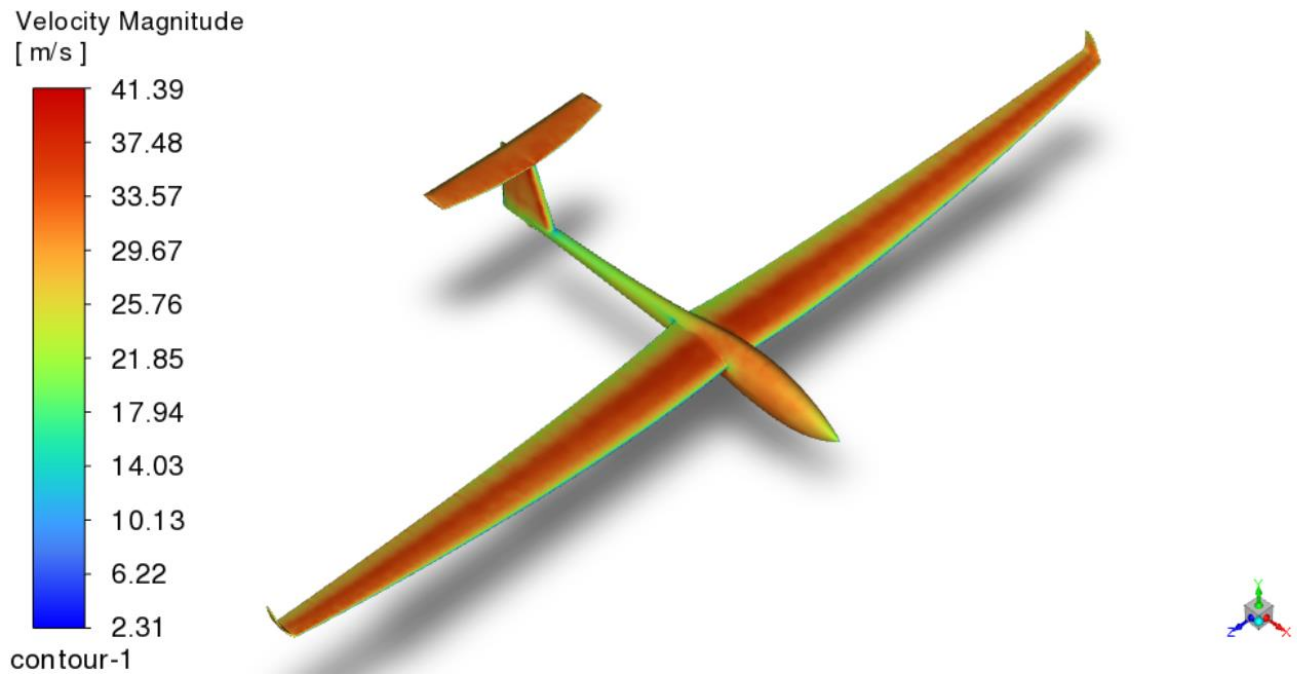


Figure 5: Velocity magnitude contour around the aerofoil profile

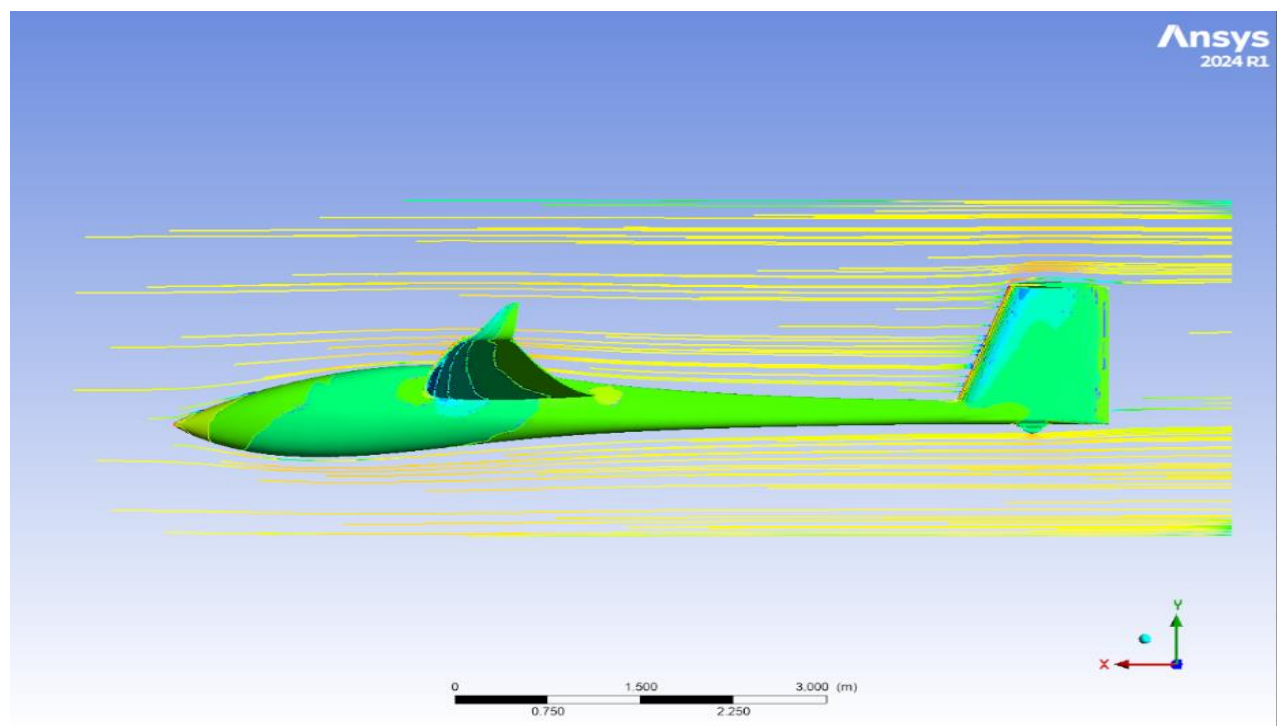


Figure 6: Velocity vectors showcasing fluid flow direction and trailing edge wake

4. Engineering Insights & Post-Processing

- **Velocity Contours:** Extracted from ANSYS Fluent, the visual results validate Bernoulli's principle, showing flow acceleration over the upper suction side of the wing and flow deceleration near the trailing edge.
- **Pressure Gradients:** The total pressure contours clearly demonstrate the lift-generation mechanism, capturing the high-pressure stagnation zone underneath the wing and the low-pressure zone on top.

Core Skills Demonstrated

- **Software Interoperability:** Smooth workflow integration between SolidWorks (CAD) and ANSYS (CAE).
- **CFD Pre-Processing & Numerical Setup:** Domain creation, boundary naming, and selecting appropriate relaxation factors and discretization schemes.
- **Mathematical Validation:** Interpreting residual plots to ensure grid independence and physical convergence before analyzing post-processed data.

What We Can Deliver for Your Project?

If you are looking to optimize your hardware, electronics enclosure, or EV battery pack, we can provide:

1. **Fluid Domain Optimization:** Minimizing pressure drops and eliminating flow recirculation/dead zones.
2. **Thermal Management Solutions:** Keeping your high power components or battery cells within optimal operating temperatures.
3. **Comprehensive Engineering Reports:** High resolution contours, vector plots, and actionable design recommendations backed by solid engineering metrics.

“Thank you for your time”

For orders, or project inquiries, please feel free to contact:

 apexenggrp@gmail.com